

Functions

ANSWER KEY



Evaluating functions

- 1 If $f(x) = 3x - 4$, find $f(5)$.
 $f(5) = 3(5) - 4 = 11$.
 - 2 If $f(x) = x^2 - 2x + 1$, find $f(-2)$.
 $f(-2) = 4 + 4 + 1 = 9$.
 - 3 If $g(x) = \sqrt{x + 5}$, find $g(4)$.
 $g(4) = \sqrt{9} = 3$.
 - 4 If $h(x) = \frac{2x}{\frac{6}{1}}$, find $h(3)$.
 $h(3) = \frac{2}{2} = 3$.
-

Domain and range

- 5 State the domain of $f(x) = \sqrt{x - 3}$.
 $x - 3 \geq 0$, so the domain is $x \geq 3$.
 - 6 State the domain of $f(x) = \frac{1}{x - 5}$.
All real numbers except $x = 5$ (where the denominator is zero).
 - 7 State the domain of $f(x) = \sqrt{7 - x}$.
 $7 - x \geq 0$, so the domain is $x \leq 7$.
 - 8 State the range of $f(x) = x^2 + 2$.
Since $x^2 \geq 0$ for all real x , the range is $f(x) \geq 2$.
-

Transformations: shifts

- 9 The graph of $y = x^2$ is shifted 3 units right and 2 units down. Write the new equation.
 $y = (x - 3)^2 - 2$.
 - 10 The graph of $y = |x|$ is shifted 4 units left and 1 unit up. Write the new equation.
 $y = |x + 4| + 1$.
 - 11 Describe the transformation that takes $y = x^2$ to $y = (x + 2)^2 - 5$.
A shift 2 units left and 5 units down.
-

Transformations: reflections and stretches

- 12 The graph of $y = x^2$ is reflected over the x -axis and stretched vertically by a factor of 3. Write the new equation.
 $y = -3x^2$.
- 13 Describe the transformation that takes $y = \sqrt{x}$ to $y = 2\sqrt{x}$.
A vertical stretch by a factor of 2.
- 14 Describe the transformation that takes $y = x^2$ to $y = -(x - 1)^2 + 4$.
A reflection over the x -axis, then a shift 1 unit right and 4 units up.
-

Composite functions

- 15 Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $(f \circ g)(2)$.
 $g(2) = 4 - 3 = 1$. $f(1) = 2(1) + 1 = 3$.
- 16 Using the same f and g , find $(g \circ f)(2)$.
 $f(2) = 5$. $g(5) = 25 - 3 = 22$.
- 17 Using the same f and g , find a general formula for $(f \circ g)(x)$.
 $f(g(x)) = 2(x^2 - 3) + 1 = 2x^2 - 6 + 1 = 2x^2 - 5$.
- 18 Using the same f and g , find a general formula for $(g \circ f)(x)$.
 $g(f(x)) = (2x + 1)^2 - 3 = 4x^2 + 4x + 1 - 3 = 4x^2 + 4x - 2$.
-

Inverse functions

- 19 Find the inverse of $f(x) = 3x - 2$.
 $y = 3x - 2 \Rightarrow x = \frac{y + 2}{3}$. So $f^{-1}(x) = \frac{x + 2}{3}$.
- 20 Find the inverse of $f(x) = \frac{x - 4}{5}$.
 $y = \frac{x - 4}{5} \Rightarrow 5y = x - 4 \Rightarrow x = 5y + 4$. So $f^{-1}(x) = 5x + 4$.
- 21 Show that $f(x) = 2x + 6$ and $g(x) = \frac{x - 6}{2}$ are inverses by computing $f(g(x))$.
 $f(g(x)) = 2\left(\frac{x - 6}{2}\right) + 6 = (x - 6) + 6 = x$, confirming they are inverses.
- 22 Find the inverse of $f(x) = x^3 + 1$.
 $y = x^3 + 1 \Rightarrow x = \sqrt[3]{y - 1}$. So $f^{-1}(x) = \sqrt[3]{x - 1}$.
-

Piecewise functions

- 23** A function is defined by $f(x) = x + 2$ for $x < 1$, and $f(x) = 3x - 1$ for $x \geq 1$. Find $f(-2)$.
Since $-2 < 1$, use the first rule: $f(-2) = -2 + 2 = 0$.
- 24** Using the same piecewise function, find $f(1)$.
Since $1 \geq 1$, use the second rule: $f(1) = 3(1) - 1 = 2$.
- 25** Using the same piecewise function, find $f(4)$.
Since $4 \geq 1$, use the second rule: $f(4) = 3(4) - 1 = 11$.
-

Even and odd functions

- 26** Determine whether $f(x) = x^2 - 3$ is even, odd, or neither.
 $f(-x) = (-x)^2 - 3 = x^2 - 3 = f(x)$, so f is even.
- 27** Determine whether $f(x) = x^3 - x$ is even, odd, or neither.
 $f(-x) = -x^3 + x = -(x^3 - x) = -f(x)$, so f is odd.
- 28** Determine whether $f(x) = x^2 + x$ is even, odd, or neither.
 $f(-x) = x^2 - x$, which equals neither $f(x)$ nor $-f(x)$, so f is neither even nor odd.
-

Increasing, decreasing, and average rate of change

- 29** Find the average rate of change of $f(x) = x^2$ from $x = 1$ to $x = 4$.
 $\frac{f(4) - f(1)}{4 - 1} = \frac{16 - 1}{3} = \frac{15}{3} = 5$.
- 30** Find the average rate of change of $f(x) = 2x^2 - 3x$ from $x = 0$ to $x = 3$.
 $f(3) = 18 - 9 = 9$, $f(0) = 0$. $\frac{9 - 0}{3} = 3$.
- 31** State the interval on which $f(x) = (x - 2)^2$ is decreasing.
The vertex is at $x = 2$ and the parabola opens upward, so f is decreasing for $x < 2$.
-

Operations on functions

- 32** Let $f(x) = x + 3$ and $g(x) = x^2 - 1$. Find $(f + g)(x)$.
 $(f + g)(x) = (x + 3) + (x^2 - 1) = x^2 + x + 2$.
- 33** Using the same f and g , find $(f \cdot g)(2)$.
 $f(2) = 5$, $g(2) = 3$. $(f \cdot g)(2) = 5 \times 3 = 15$.
- 34** Using the same f and g , find $(f/g)(3)$.
 $f(3) = 6$, $g(3) = 8$. $(f/g)(3) = \frac{6}{8} = \frac{3}{4}$.

Reading a function graph

35 The graph of $y = f(x) = x^2 - 4$ is shown.

- a State the zeros of f . $x = -2$ and $x = 2$ (where the curve crosses the x -axis).
- b State the domain and range of f . Domain: all real numbers. Range: $y \geq -4$.

Word problems: functions in context

36 This question has two parts. A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 150$.

a Evaluate $P(10)$ to find the profit from selling 10 units.

$$P(10) = -2(100) + 800 - 150 = -200 + 800 - 150 = 450 \text{ dollars.}$$

b Find the number of units x that maximizes profit (using the vertex of the parabola), and state the maximum profit. The vertex is at $x = -\frac{80}{2(-2)} = 20$ units. $P(20) = -2(400) + 1600 - 150 = 650$ dollars — the maximum profit.

37 This question has two parts. The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9}{5}C + 32$. Meanwhile, the temperature in a city t hours after 6 a.m. is modeled by $C(t) = -0.5t^2 + 6t + 10$ (in degrees Celsius), for $0 \leq t \leq 12$.

a Find $C(4)$, the Celsius temperature 4 hours after 6 a.m.

$$C(4) = -0.5(16) + 24 + 10 = -8 + 24 + 10 = 26^\circ\text{C.}$$

b Find $(F \circ C)(4)$, the Fahrenheit temperature at that time.

$$(F \circ C)(4) = \frac{9}{5}(26) + 32 = 46.8 + 32 = 78.8^\circ\text{F.}$$

MCQ — Evaluating functions

38 If $f(x) = 4x - 7$, find $f(6)$.

a. 17 ✓

b. 24

c. 31

d. 10

$$4(6) - 7 = 24 - 7 = 17.$$

39 If $f(x) = x^2 - 3x + 2$, find $f(-3)$.

a. 2

b. -16

c. 20 ✓

d. 16

$$(-3)^2 - 3(-3) + 2 = 9 + 9 + 2 = 20.$$

40 If $g(x) = \sqrt{x + 9}$, find $g(7)$.

a. 8

b. 16

c. 4 ✓

d. $\sqrt{16}$

$$g(7) = \sqrt{16} = 4.$$

41 If $h(x) = \frac{3x}{x-2}$, find $h(5)$.

a. 15

b. 5 ✓

c. 3

d. 1.5

$$h(5) = \frac{15}{3} = 5.$$

42 If $f(x) = 2x^2 + 1$, find $f(3) - f(1)$.

a. 18

b. 8

c. 16 ✓

d. 22

$$f(3) = 19, f(1) = 3; 19 - 3 = 16.$$

MCQ — Domain and range

43 State the domain of $f(x) = \sqrt{x - 5}$.

- a. $x > 5$
- b. $x \geq 5$ ✓
- c. $x \leq 5$
- d. $x \neq 5$

The expression under the root must be ≥ 0 : $x - 5 \geq 0 \Rightarrow x \geq 5$.

44 State the domain of $f(x) = \frac{1}{x - 7}$.

- a. All reals except $x = 7$ ✓
- b. $x > 7$
- c. $x \geq 7$
- d. All real numbers

The denominator cannot be 0, so $x \neq 7$.

45 State the domain of $f(x) = \sqrt{9 - x}$.

- a. $x \leq 9$ ✓
- b. $x < 9$
- c. $x \neq 9$
- d. $x \geq 9$

$9 - x \geq 0 \Rightarrow x \leq 9$.

46 State the range of $f(x) = x^2 + 5$.

a. $y > 5$

b. All real numbers

c. $y \leq 5$

d. $y \geq 5$ ✓

Since $x^2 \geq 0$, the minimum value of f is 5: range $y \geq 5$.

47 The graph shows $f(x)$ defined only where plotted. State its domain.

a. $[-3, 3]$ ✓

b. All real numbers

c. $[-4, 4]$

d. $[0, 3]$

The curve is only plotted for x between -3 and 3 .

MCQ — Transformations: shifts

48 The graph of $y = x^2$ is shifted 5 units right and 3 units down. Write the new equation.

a. $y = (x - 5)^2 + 3$

b. $y = (x - 5)^2 - 3$ ✓

c. $y = (x + 5)^2 - 3$

d. $y = (x - 3)^2 - 5$

Right 5: replace x with $x - 5$; down 3: subtract 3.

49 The graph of $y = |x|$ is shifted 2 units left and 4 units up. Write the new equation.

a. $y = |x + 4| + 2$

b. $y = |x + 2| + 4$ ✓

c. $y = |x + 2| - 4$

d. $y = |x - 2| + 4$

Left 2: replace x with $x + 2$; up 4: add 4.

50 Describe the transformation that takes $y = x^2$ to $y = (x - 3)^2 + 2$.

a. Shift right 2, up 3

b. Shift right 3, down 2

c. Shift left 3, up 2

d. **Shift right 3, up 2** ✓

Replacing x with $x - 3$ shifts right 3; adding 2 shifts up 2.

51 The graph of $y = \sqrt{x}$ is shifted 6 units right. Write the new equation.

a. $y = \sqrt{x + 6}$

b. $y = \sqrt{x} - 6$

c. $y = \sqrt{x} + 6$

d. $y = \sqrt{x - 6}$ ✓

Right 6: replace x with $x - 6$.

MCQ — Transformations: reflections and stretches

- 52 The graph of $y = x^2$ is reflected over the x -axis and stretched vertically by a factor of 2. Write the new equation.

a. $y = -\frac{1}{2}x^2$

b. $y = (-2x)^2$

c. $y = -2x^2$ ✓

d. $y = 2x^2$

Reflect over x -axis: negate; stretch by 2: multiply by 2.

- 53 Describe the transformation that takes $y = \sqrt{x}$ to $y = 3\sqrt{x}$.

a. Reflection over the x -axis

b. Vertical shift up 3

c. **Vertical stretch by a factor of 3** ✓

d. Horizontal stretch by a factor of 3

Multiplying the output by 3 stretches the graph vertically by a factor of 3.

- 54 Describe the transformation that takes $y = x^2$ to $y = -(x + 2)^2 + 1$.

a. Reflect over x -axis, shift right 2, down 1

b. Reflect over y -axis, shift right 2, up 1

c. **Reflect over x -axis, shift left 2, up 1** ✓

d. Shift left 2, up 1 (no reflection)

The negative sign reflects over the x -axis; $+2$ shifts left 2; $+1$ shifts up 1.

- 55 The graph of $y = \sqrt{x}$ is reflected over the y -axis. Write the new equation.

a. $y = -\sqrt{-x}$

b. $y = -\sqrt{x}$

c. $y = \sqrt{-x}$ ✓

d. $y = \sqrt{x}$

Reflecting over the y -axis replaces x with $-x$: $y = \sqrt{-x}$.

MCQ — Composite functions

56 Let $f(x) = 3x + 2$ and $g(x) = x^2 - 1$. Find $(f \circ g)(3)$.

a. 26 ✓

b. 23

c. 11

d. 29

$$g(3) = 8; f(8) = 3(8) + 2 = 26.$$

57 Using the same f and g , find $(g \circ f)(3)$.

a. 100

b. 120 ✓

c. 121

d. 26

$$f(3) = 11; g(11) = 11^2 - 1 = 120.$$

58 Using the same f and g , find a general formula for $(f \circ g)(x)$.

a. $3x^2 - 1$ ✓

b. $9x^2 - 1$

c. $3x^2 - 3$

d. $3x^2 + 2$

$$f(g(x)) = 3(x^2 - 1) + 2 = 3x^2 - 1.$$

59 Using the same f and g , find a general formula for $(g \circ f)(x)$.

a. $9x^2 + 12x + 3$ ✓

b. $3x^2 + 2x - 1$

c. $9x^2 + 4$

d. $9x^2 + 12x - 1$

$$g(f(x)) = (3x + 2)^2 - 1 = 9x^2 + 12x + 4 - 1 = 9x^2 + 12x + 3.$$

60 Let $f(x) = x + 4$ and $g(x) = 2x$. Find $(f \circ g)(5)$.

a. 10

b. 9

c. 14 ✓

d. 18

$$g(5) = 10; f(10) = 14.$$

MCQ — Inverse functions

61 Find the inverse of $f(x) = 4x - 3$.

a. $f^{-1}(x) = \frac{x-3}{4}$

b. $f^{-1}(x) = \frac{x+4}{3}$

c. $f^{-1}(x) = \frac{x+3}{4}$ ✓

d. $f^{-1}(x) = 4x + 3$

$$y = 4x - 3 \Rightarrow x = \frac{y+3}{4}.$$

62 Find the inverse of $f(x) = \frac{x-7}{2}$.

a. $f^{-1}(x) = 2x - 7$

b. $f^{-1}(x) = 2(x - 7)$

c. $f^{-1}(x) = 2x + 7$ ✓

d. $f^{-1}(x) = \frac{x+7}{2}$

$$y = \frac{x-7}{2} \Rightarrow x = 2y + 7.$$

63 Show that $f(x) = 3x + 9$ and $g(x) = \frac{x-9}{3}$ are inverses by computing $f(g(x))$.

a. x ✓

b. $x + 9$

c. $3x$

d. $x - 9$

$$f(g(x)) = 3\left(\frac{x-9}{3}\right) + 9 = x - 9 + 9 = x.$$

64 Find the inverse of $f(x) = x^3 - 2$.

a. $f^{-1}(x) = \sqrt[3]{x} + 2$

b. $f^{-1}(x) = (x + 2)^3$

c. $f^{-1}(x) = \sqrt[3]{x-2}$

d. $f^{-1}(x) = \sqrt[3]{x+2}$ ✓

$$y = x^3 - 2 \Rightarrow x = \sqrt[3]{y+2}.$$

65 Find the inverse of $f(x) = 5x + 10$.

a. $f^{-1}(x) = \frac{x-10}{5}$ ✓

b. $f^{-1}(x) = \frac{x+10}{5}$

c. $f^{-1}(x) = \frac{x}{5} - 10$

d. $f^{-1}(x) = 5x - 10$

$$y = 5x + 10 \Rightarrow x = \frac{y-10}{5}.$$

MCQ — Piecewise functions

66 A function is defined by $f(x) = x + 3$ for $x < 2$, and $f(x) = 2x - 1$ for $x \geq 2$. Find $f(-3)$.

a. -7

b. 0 ✓

c. 6

d. 1

$$x = -3 < 2: f(-3) = -3 + 3 = 0.$$

67 Using the same piecewise function, find $f(2)$.

a. 3 ✓

b. 4

c. 0

d. 5

$$x = 2 \geq 2: f(2) = 2(2) - 1 = 3.$$

68 Using the same piecewise function, find $f(5)$.

a. 5

b. 9 ✓

c. 7

d. 8

$$x = 5 \geq 2: f(5) = 2(5) - 1 = 9.$$

69 A function is defined by $f(x) = 2x$ for $x \leq 0$, and $f(x) = x^2 + 1$ for $x > 0$. Find $f(0) + f(3)$.

a. 1

b. 11

c. 0

d. 10 ✓

$f(0) = 0$; $f(3) = 9 + 1 = 10$; sum = 10.

MCQ — Even and odd functions

70 Determine whether $f(x) = x^2 - 7$ is even, odd, or neither.

a. Neither

b. Even ✓

c. Odd

d. Both even and odd

$f(-x) = x^2 - 7 = f(x)$: even.

71 Determine whether $f(x) = x^3 - 4x$ is even, odd, or neither.

a. Neither

b. Both even and odd

c. Even

d. Odd ✓

$f(-x) = -x^3 + 4x = -f(x)$: odd.

72 Determine whether $f(x) = x^2 + 3x$ is even, odd, or neither.

a. Both even and odd

b. Neither ✓

c. Odd

d. Even

$f(-x) = x^2 - 3x$, which is neither $f(x)$ nor $-f(x)$: neither.

73 The graph shown is symmetric about the y -axis. Is the function even, odd, or neither?

a. Odd

b. Neither

c. Cannot be determined

d. Even ✓

Symmetry about the y -axis is exactly the definition of an even function.

MCQ — Increasing, decreasing, and average rate of change

74 Find the average rate of change of $f(x) = x^2$ from $x = 2$ to $x = 5$.

a. 3

b. 7 ✓

c. 21

d. 29

$$\frac{f(5) - f(2)}{5 - 2} = \frac{25 - 4}{3} = 7.$$

75 Find the average rate of change of $f(x) = 3x^2 - 2x$ from $x = 0$ to $x = 4$.

a. 40

b. 8.5

c. 10 ✓

d. 46

$$f(4) = 40, f(0) = 0; \text{rate} = \frac{40}{4} = 10.$$

76 State the interval on which $f(x) = (x - 3)^2$ is decreasing.

a. $(-\infty, 3)$ ✓

b. $(-\infty, -3)$

c. $(-3, \infty)$

d. $(3, \infty)$

The parabola's vertex is at $x = 3$; it decreases for $x < 3$.

77 Find the average rate of change of $f(x) = x^3$ from $x = 1$ to $x = 3$.

a. 8

b. 26

c. 9

d. 13 ✓

$$f(3) = 27, f(1) = 1; \text{rate} = \frac{26}{2} = 13.$$

MCQ — Operations on functions

78 Let $f(x) = x + 5$ and $g(x) = x^2 - 2$. Find $(f + g)(x)$.

a. $x^2 - x + 3$

b. $x^2 + x + 3$ ✓

c. $x^2 + x + 7$

d. $x^2 + 2x + 3$

$$(x + 5) + (x^2 - 2) = x^2 + x + 3.$$

79 Using the same f and g , find $(f \cdot g)(3)$.

a. 56 ✓

b. 48

c. 62

d. 50

$$f(3) = 8, g(3) = 7: 8 \times 7 = 56.$$

80 Using the same f and g , find $(f/g)(4)$.

a. $\frac{7}{14}$

b. $\frac{9}{16}$

c. $\frac{9}{14}$ ✓

d. $\frac{14}{9}$

$$f(4) = 9, g(4) = 14: \frac{9}{14}.$$

81 Let $f(x) = 2x$ and $g(x) = x + 6$. Find $(f - g)(x)$.

a. $3x + 6$

b. $x - 6$ ✓

c. $x + 6$

d. $-x - 6$

$$2x - (x + 6) = x - 6.$$

MCQ — Reading a function graph

82 The graph shows $y = f(x) = x^2 - 9$. Find $f(3)$ by reading the graph.

a. -9

b. 6

c. 9

d. 0 ✓

The curve crosses the x -axis at $x = 3$, so $f(3) = 0$.

83 Using the same graph, find the values of x where $f(x) = 0$ (the x -intercepts).

a. $x = 0$ and $x = 9$

b. $x = -9$ and $x = 9$

c. $x = 3$ only

d. $x = -3$ and $x = 3$ ✓

The curve crosses the x -axis at $x = -3$ and $x = 3$.

84 Using the same graph, find the y -intercept.

a. -3

b. 9

c. 0

d. -9 ✓

At $x = 0$, the curve crosses the y -axis at $y = -9$.

MCQ — Word problems: functions in context

- 85 A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 300$. Find the profit when 15 units are sold.

a. 450 dollars ✓

b. 150 dollars

c. 600 dollars

d. 900 dollars

$$P(15) = -2(225) + 1,200 - 300 = 450 \text{ dollars.}$$

- 86 The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9C}{5} + 32$. Find $F(20)$.

a. 72°F

b. 68°F ✓

c. 36°F

d. 52°F

$$F(20) = \frac{9(20)}{5} + 32 = 36 + 32 = 68.$$

- 87 A ball's height (in meters) t seconds after being thrown is $h(t) = -5t^2 + 20t + 2$. Find the height at $t = 2$ seconds.

a. 2

b. 18

c. 22 ✓

d. 42

$$h(2) = -5(4) + 40 + 2 = 22 \text{ meters.}$$